Statement of Interest for HEARTS Workshop at CHI ’26

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ACM Reference Format:

Anirban Mukhopadhyay. 2026. Statement of Interest for HEARTS Workshop at CHI ’26. 1, 1 (February 2026), 2 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Motivation Statement

I am motivated to attend this workshop because my research sits at the intersection of human–AI collaboration, collaborative sensemaking, and cybersecurity investigations. In my recent work, I have explored how AI-augmented systems can support teams conducting Open Source Intelligence (OSINT) investigations, particularly in educational and training contexts where learners engage in real-world vulnerability assessment tasks [1, 3, 4]. Through projects such as OSINT Clinic, I investigate how collaborative workflows, transparency mechanisms, and proactive AI assistance can help teams reason through complex investigative tasks while maintaining ethical and operational safeguards.

This workshop offers a valuable opportunity to engage with the growing conversation around red teaming as both a technical and collaborative practice. I am particularly interested in how red-teaming methods can be adapted into structured learning environments where students and novice analysts develop investigative expertise while navigating uncertainty and risk. Given my background in designing sociotechnical systems and AI agents that proactively support teamwork, I hope to contribute perspectives on how human-centered design principles can make red-teaming practices more accessible, responsible, and scalable for broader educational and research communities.

2 Topics That I Want To Discuss

If accepted, I hope to discuss how red teaming can be framed as a collaborative learning and sensemaking activity rather than solely a technical exercise. Specifically, I am interested in exploring how Generative AI and agentic systems can support red-team workflows by helping teams plan investigations, coordinate roles, surface alternative hypotheses, and reflect on findings without replacing critical human judgment.

I am also eager to engage in conversations about safety, ethics, and labor within red-teaming environments. My research has shown that investigative work can involve cognitive overload, uncertainty, and potential exposure to harmful content; thus, I am interested in discussing how system design and AI agents can help safeguard practitioners while maintaining rigor. Another topic I would like to explore is how collaborative AI agents—such as facilitator-style or peer-like agents—can scaffold group processes during adversarial evaluation or vulnerability assessment tasks, supporting coordination and decision-making while preserving team autonomy.

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3 Related Prior Work

My research background combines CSCW, human-centered AI, and cybersecurity investigation workflows. Through the OSINT Clinic project [1], I designed and evaluated AI-augmented collaborative environments that help undergraduate student teams conduct vulnerability assessments using a multi-tool investigative workflow. This work explores how AI support can assist with planning, information gathering, and synthesis while ensuring transparency and user control. Related projects such as OSINT Research Studios [3] and CoSINT [4] further examine how crowdsourcing and structured collaboration can scale investigative work while reducing duplication and information silos.

In parallel, my work on proactive generative AI agents investigates how agent roles influence team collaboration and performance under time pressure [2]. By designing facilitator and peer-style AI agents that actively support group coordination, my research identifies design considerations for integrating AI into team processes rather than treating AI as a passive assistant. I see strong overlap between this work and red-teaming contexts, where teams must coordinate rapidly, challenge assumptions, and reason adversarially.

4 Future Directions

From a CSCW perspective, I believe the field should prioritize understanding red teaming as a socio-technical practice that emerges through collaboration, shared accountability, and human judgment. Much of the existing discourse focuses on tools and technical outcomes, while less attention is paid to team dynamics, coordination strategies, and how expertise develops in collaborative adversarial settings.

Future research should also examine how AI systems can responsibly augment red-teaming work. This includes understanding when AI agents should proactively intervene, how transparency and verifiability can be incorporated into AI recommendations, and how to design safeguards that prevent misuse or over-reliance. Finally, I believe CSCW should explore how to democratize red-teaming practices by creating learning-oriented frameworks that enable students and early-career practitioners to participate safely and effectively.

References

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